What Fine-tuning changes

Pretraining teaches a model broad patterns from substantial amounts of text. Instruction tuning makes it more responsive to requests. Our supervised fine-tuning, or SFT, adds examples of a narrower task, adjusting weights so desired responses become more likely for similar inputs. The model does not acquire a separate searchable filing cabinet; what it learns becomes distributed across parameters.

Consider a small electronics retailer. The assistant must convert customer messages into JSON containing a category, urgency and short next action. A system prompt may get most examples right, but unusual messages cause inconsistent field names or wandering explanations. Repeated, carefully labelled examples can help. That is a behaviour problem. If the same retailer asks for today's stock position, the correct answer belongs in the inventory system, reached through a tool. If it asks for the latest warranty terms, use the RAG pipeline we built previously.

Before training, collect fifty realistic unseen prompts and run the base model with your best instructions. Save every output. Without this baseline, there is no honest way to know whether an expensive training exercise improved anything.

Choose the smallest intervention

Try prompting first. Put the required schema in a system instruction, supply two contrasting examples, and request an explicit refusal when required information is absent. If this works consistently, stop. A reliable prompt is cheaper to modify than an adapter.

Try retrieval when factual information lives in documents. Try deterministic Python code when the task is arithmetic, field validation or an exact formatting conversion that does not require interpretation. Use fine-tuning when many labelled examples reveal a persistent pattern that instructions alone cannot fix: categorisation, specialist terminology, house-style responses or structured extraction.

Fine-tuning can also introduce regression. A narrowly trained assistant may become overconfident, lose some general abilities or imitate erroneous labels. Fine-tuning on private customer records can create memorisation and disclosure risks. Keep records anonymised, respect licensing, and do not treat an adapter as a security boundary. If the proposed job still requires privileged knowledge, control that access outside the model.

Full Fine-Tuning, LoRA and QLoRA

In full fine-tuning, training updates a large portion or all of the model's parameters. That consumes substantial memory because optimisation retains gradients and other state as well as model weights. For our personal workstation project, full fine-tuning is unnecessary.

Low-Rank Adaptation, or LoRA, takes a lighter route. It freezes the original weights and learns small additional matrices in selected layers. During inference, those matrices modify how the base model behaves. The result is a compact adapter rather than another complete multi-gigabyte model. Hugging Face's PEFT library provides the relevant machinery, including saving adapters and switching between them.

Think of a base model as a camera body and an adapter as a lens selected for a particular assignment. One base could support a support-ticket adapter and a product-description adapter, provided each was trained and evaluated separately.

